Designing phase-controlled robust state-dependent kicks for trapped-ion quantum computing

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ABSTRACT

Quantum computers offer a unique capability to model many-body systems, with potential applications that could transform the pharmaceutical industry. Trapped-ion systems are a central research focus that feature long coherence times and have already demonstrated low infidelity quantum logic gates. Here, we study the non-adiabatic (fast) regime of gate operations in which we employ state-dependent kicks (SDKs) to implement a gate, leading to a significantly reduced gate time. Fast gates inherit compounding SDK errors as they are susceptible to experimental noise, and as such we require low infidelity SDKs to be robust. This study introduces phase control into our robust SDK scheme to enable geometric cancellation of errors that arise primarily due to laser noise. We use numerical methods to find optimised pulse parameters that minimise SDK infidelity in the presence of noise. Our results demonstrate that the infidelity of phase-controlled robust SDKs can achieve up to two orders of magnitude lower infidelity under laser timing jitter compared with the benchmark scheme. These findings demonstrate that phase control can significantly enhance the robustness of SDKs, representing a meaningful step toward scalable and robust trapped-ion quantum computing.

1. INTRODUCTION

Scientific and commercial interest in quantum computers (QCs) is driven by their proposed abilities to model complex chemical structures which could potentially transform the pharmaceutical R&D market that is worth hundreds of billions of US dollars (Evers et al. 2021). This is because QCs exploit quantum phenomena, such as entanglement and superposition (Monroe 2002), allowing them to model many-body systems whose Hilbert space dimension grows exponentially, rendering classical simulation infeasible. In particular, previously impossible quantum chemistry simulations would allow for a rapid development of drug design (Blunt et al. 2022).

For a QC to be useful, it must employ quantum bits (qubits), which are two-level systems and can be expressed in the spin basis as a superposition of the states $|1\rangle$ and $|0\rangle$ (Nielsen & Chuang 2010). Qubits require the property of having their final state, after a computation starting from a well-defined input state, to be measurable. Additionally, much like classical computers, a QC requires the functionality to perform operations on the qubits via a quantum logic gate. Unitary operations are used to enable this, and the quality of the operation is given by its fidelity, $\mathcal{F}$, which measures the overlap between a gate's ideal operation on a quantum state and its actual result.

There are many avenues that one may explore to produce a successful QC. The most promising ones are superconducting qubits (Chen et al. 2014; Arute et al. 2019; Kjaergaard et al. 2020), trapped ions (Home et al. 2009; Moses et al. 2023; Chen et al. 2024) and neutral atoms (Tamrakar & Sharma 2019; Graham et al. 2022). In this paper, we study trapped ions due to the larger (two orders of magnitude) ratio of decoherence time to the duration of the gates (Schäfer et al. 2018; Wang et al. 2021). This is done with a two-qubit entangling gate fidelity of 99.94% (Clark et al. 2021) and 99.97% (Löschnauer et al. 2024); single-qubit gates are implemented with a 99.999% fidelity (Löschnauer et al. 2024).

In trapped-ion systems, gates are achieved by coupling the spin state ($|0\rangle$ or $|1\rangle$) with motional modes. These gates are operated in two regimes: adiabatic (in the quantum sense) and non-adiabatic (fast). Adiabatic gates 'slowly' (longer than the motional period) perform operations on the ions ensuring that they remain an instantaneous eigenstate of the Hamiltonian. This results in a robust process that is tolerant to ambient noise and insensitive to fluctuations in

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experimental parameters (Zhang et al. 2009). The principal limitation in the adiabatic regime is that one must spectrally resolve individual motional-sideband transitions, which forces a gate time that is long compared with the motional period of the ion (Schäfer et al. 2018). To resolve this, fast gates have been proposed (García-Ripoll et al. 2003, 2005; Zhu et al. 2006; Wong-Campos et al. 2017) where the adiabatic condition is relaxed and all motional sidebands are simultaneously driven. A sequence of qubit-state-dependent kicks (SDKs) can be designed using lasers to entangle the motional and spin states of the ion. A sequence of SDKs with numerically-optimized timings disentangles the spin and motion of the ions by the end of the gate, while generating entanglement between the two addressed qubits. SDKs are short (pico to nanosecond time-scale) laser-driven momentum pulses whose direction depends on the qubit state. This, therefore, results in the same outcome as adiabatic gates but with a significantly (an order of magnitude) improved gate duration (Wong-Campos et al. 2017; Schäfer et al. 2018).

The cost of this improvement is typically a reduction in the gate's robustness to noise. Specifically, it has previously been shown that errors in individual SDKs compound with the number of SDKs in a given gate (Sidorov et al. 2021). The noise sources that impact the SDKs (which make up the gate sequence) are the following: systematic drifts and random fluctuations in laser intensity, timing jitter in the repetition rate of the laser and beatnote noise in the atom state transition due to the SDK. Our collaborators have studied the sensitivity of SDKs due to these noises and demonstrated that low infidelities ($< 10^{-6}$) can be maintained; however these solutions are not robust.

In this work, we propose a new approach to reducing noise sensitivity by designing phase-controlled robust SDKs that introduce an additional degree of freedom: the phase ϕ of each laser pulse that constitutes an SDK. This is therefore an extension of work by (Mizrahi et al. 2013b) which only considered pulse timing optimisation. More recently, optimisation of SDK amplitude modulation has been studied by our industry collaborators and hence, our work extends this via phase modulation. The motivation for this arises from well-known pulse sequences in NMR (Tycko et al. 1985; Wimperis 1994) which take advantage of laser phase. Additionally, phase modulation has been shown to lead to adiabatic gates that are robust to static and time-varying errors (Milne et al. 2020) which we aim to apply in the fast regime as well. Phase controls allow a change of direction when traversing the Bloch sphere, permitting the geometric cancellation of errors that occur as a result of noise. We model noise on a particular pulse parameter by sampling Gaussian-distributed fluctuations. We use numerical methods to optimise pulse parameters such that SDKs are robust to noise and maintain their fidelity, $\mathcal{F}$. The SDK model uses δ-shaped pulses according to (Mizrahi 2013) that also allow for the phase, ϕ, of each pulse in a SDK to be controlled. A δ pulse is defined by its picosecond duration which is several orders of magnitude shorter than the microsecond timescale of the ion motion, allowing the ion to be treated as stationary during the pulse. The additional phase degree of freedom is used with the aim of further increasing the fidelity of the SDKs which comes at the cost of increasing the dimensionality of the optimisation space. To calculate the success of each SDK we use a cost function that evaluates the infidelity (defined as $1 - \mathcal{F}$) for each noise realisation from the Gaussian distribution and returns the average infidelity, $\langle 1 - \mathcal{F} \rangle$, for a given noise tolerance. Minimising $\langle 1 - \mathcal{F} \rangle$ yields the robust SDK control parameters.

This paper is organised as follows. In Section 2 we outline the implementation of the SDK theory to model robust SDKs and the associated numerical methods used for optimising pulse parameters. Section 3 contains our main results, including visualisations of the pulses and the population of the states after an SDK sequence. We also assess our robust SDK designs by comparing the $\langle 1 - \mathcal{F} \rangle$ under the listed noise sources to show the improvement by benchmarking against noise-free SDKs. A summary, along with future directions of this work are given in Section 4.

2. METHODS

In this section we present our numerical modelling and optimisation approach for SDK sequences. We focus on minimising gate infidelity under realistic noise models, and we provide an overview of our simulation framework and code infrastructure.

2.1. SDK scheme

Quantum gates are realised by unitary evolutions, therefore we require our SDKs to be composed of unitary operators. Fast gates rely on state-dependent momentum transfer, with the ideal SDK unitary given by the following in terms of the Pauli spin matrices ($\hat{\sigma}_x$ and $\hat{\sigma}_z$), and the position operator $\hat{x}$ (Savill-Brown et al. 2025),

$$\hat{U}_{\mathrm{SDK}} = \hat{\sigma}_x \exp\left[i\left(k\hat{x} + \phi\right)\hat{\sigma}_z\right], \tag{1}$$

where k is the wave number and ϕ is the phase of the laser. However, implementing the action of an SDK via Eq. (1) in a numerical model is not straightforward. This is because to achieve a perfect population inversion (e.g. $|0\rangle \rightarrow |1\rangle$) our pulse duration, τ, must satisfy $\tau\omega_{\mathrm{HF}} \ll 1$. That is, the product of the pulse duration and the hyperfine qubit transition frequency (typically on the order of 10 GHz) must be much less than one which is not possible for our case of $\tau \sim 10$ ps (Putnam et al. 2024). By failing to meet this criterion, we excite off-resonant transitions due to the broadness of our pulse in frequency space (given the ~picosecond duration). We resolve this by introducing a pulse train to cancel out the off-resonant processes while coherently accumulating population in the target momentum order (Mizrahi 2013).

To develop this, we only consider the atom-light interaction Hamiltonian for a single ion as the pulse duration is on the order of hundreds of picoseconds whereas the motional period is on microsecond scales meaning that the ion is assumed stationary during an SDK, (Mizrahi 2013; Savill-Brown et al. 2025),

$$\hat{H} \approx \hbar\Omega(t)\cos(k\hat{x} - \omega_{\mathrm{A}}t + \phi)\hat{\sigma}_x, \tag{2}$$

$$= \hbar\Omega(t)\cos(\eta(\hat{a} + \hat{a}^\dagger) - \omega_{\mathrm{A}}t + \phi)\hat{\sigma}_x, \tag{3}$$

where $\Omega(t) \rightarrow \theta\delta(t)$ is the time-dependent Rabi frequency in the case of δ-shaped pulse with $\theta = \int \Omega(t)\mathrm{d}t$ is the pulse area and ω_{A} is the SDK beatnote. Here, we have also expressed the Hamiltonian in terms of the creation, $\hat{a}^\dagger$, and annihilation, $\hat{a}$, operators such that $\hat{x} = x_0(\hat{a} + \hat{a}^\dagger)$, in the process defining the Lamb-Dicke parameter $\eta := kx_0 = k\sqrt{\hbar/(2m\omega)}$ which measures the strength of the coupling between spin and motional states of the ion, with ω the trap frequency. An immediate observation of Eq. (2) is that we have the following spatial translation symmetry, $\hat{x} \mapsto \hat{x} + \frac{2\pi}{k}$ of the Hamiltonian. This implies that if we work in the coherent state picture defined as,

$$|\alpha\rangle = \exp(-|\alpha|^2/2)\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}|n\rangle, \tag{4}$$

then the Hamiltonian can evolve the state between coherent states $|\alpha\rangle$ and $|\alpha + in\eta\rangle$ with $n \in \mathbb{Z}$ which significantly reduces the Hilbert space, resulting in feasible compute times for numerical optimisations of state evolution. Therefore, a state in our system is described by $|0, \alpha + in\eta\rangle := |0\rangle \otimes |\alpha + in\eta\rangle$ where the $|0\rangle$ could be either spin state ($|0\rangle$ or $|1\rangle$).

In our scheme, an $M-$pulse SDK sequence consists of M instantaneous pulses, each followed by a period of free evolution (precession) of the ion system, thereby modelling the combined spin–motion dynamics. Following the derivation and summary of the results given in (Mizrahi 2013; Savill-Brown et al. 2025) one can find the unitary for a δ-shaped pulse through the Hamiltonian given in Eq. (2). This is in terms of the n-th order Bessel function of the first kind, $J_n(\theta)$ and the coherent state displacement operator $\hat{D}(\alpha) = \exp(\alpha\hat{a}^\dagger - \alpha^*\hat{a})$ to be,

$$\hat{U}_{\mathrm{p}}(\theta, \phi) = \sum_{n=-\infty}^{\infty} i^n J_n(\theta)\exp\left[i(\omega_{\mathrm{A}}t + n\phi)\right]\hat{D}(in\eta)\hat{\sigma}_x^n. \tag{5}$$

The functional form of the pulse unitary is considerably more complex than that of the overall SDK. By combining multiple pulses with optimised control parameters, these unwanted motional states destructively interfere leaving the

desired target state maximally occupied. Computationally, we truncate Eq. (5) to only include terms in the sum between $n_{\max} = \pm 8$ as motional states beyond these are not significantly populated and can hence be ignored.

The free precession of the hyperfine qubit (assuming negligible spin–motion coupling during the interval between pulses) is diagonal in the truncated basis $n = -n_{\max}, \ldots, n_{\max}$ with its entries given by,

$$\langle n|\hat{U}_{\mathrm{f}}|n\rangle = \exp\left(i\frac{\omega_{\mathrm{HF}}\mathrm{d}t}{2}(-1)^n\right). \tag{6}$$

Here, ω_{HF} is the hyperfine qubit transition frequency and $\mathrm{d}t$ is the laser timing between two pulse unitaries. The laser timing, $\mathrm{d}t$, is an important control parameter that is optimised in our code and is on the order of a nanosecond, about 3 orders of magnitude smaller than the period of micromotion (driven oscillatory motion of a trapped ion at the trap's drive frequency) (Berkeland et al. 1998). Therefore, such considerations of the motion of the ion are not required, and the free precession unitary is sufficient.

With both the free precession unitary, and pulse unitary discussed, we combine these to form the overall SDK that is implemented in our code (Mizrahi 2013),

$$\hat{U} = \prod_{i=1}^{M}\Big[\hat{U}_{\mathrm{f}}(\omega_{\mathrm{HF}}, \mathrm{d}t, n_{\max})\ \hat{U}_{\mathrm{p}}(\theta_i, \phi_i, \omega_{\mathrm{A}}, i\,\mathrm{d}t)\Big], \tag{7}$$

with all $2M + 2$ control parameters $\{\theta_i, \phi_i, \mathrm{d}t, \omega_{\mathrm{A}}\}$ being optimised so that our infidelity is minimised. Importantly, it is clear from the definition that the pulse area and the phase are unique for each pulse whilst the timing between each pulse is uniform across pulses and is itself a control parameter that we optimise.

2.2. Noise implementation

With the SDK scheme established in Section 2.1, we define the technique we use for modelling noise in our system. Given a qubit in the initial state $|\psi_{\mathrm{init}}\rangle = |0, \alpha\rangle$, upon acting $\hat{U}$ (as defined in Eq. (7)) we obtain the final state $|\psi_{\mathrm{fin}}\rangle = \hat{U}|\psi_{\mathrm{init}}\rangle$ which will contain a superposition of states of the form $|1, \alpha + in\eta\rangle$ where $|n| \leq n_{\max}$ and α is a coherent state. In the case of unoptimised control parameters, the unwanted motional modes do not cancel out because we require an optimised set of control parameters for this to occur. We can measure the success (fidelity) of our SDK by computing the overlap between $|\psi_{\mathrm{fin}}\rangle$ and the desired, target state after the SDK, $|\psi_{\mathrm{tar}}\rangle$ which is $|1, \alpha + i\eta\rangle$ as,

$$\mathcal{F} = |\langle\psi_{\mathrm{tar}}|\psi_{\mathrm{fin}}\rangle|^2. \tag{8}$$

From this, the infidelity is defined as $1 - \mathcal{F}$. We use Python to make use of the Nelder–Mead algorithm implemented in SciPy to minimise infidelity and optimal control parameters are returned such that the SDK destructively cancels the unwanted final states. Hence, if we want to measure the impact of noise in our SDK unitary (i.e. the laser pulses), we can compute the infidelity and compare this to the noise-free, 'optimal' case.

However, this is just the case for one noise realisation. In practice, we want to design robust SDKs; this means they should maintain a low infidelity even in the presence of noise realisations within a certain tolerance. Therefore we require many noise samples, which are taken from a normal distribution, $\mathcal{N}(0, \sigma^2)$, where σ is the noise tolerance (i.e. standard deviation of the distribution). The noise is then added to the control parameters we wish to make robust (pulse areas, laser timing and SDK beatnote) and Eq. (8) is evaluated for a given amount of samples from $\mathcal{N}(0, \sigma^2)$ (we typically use ~ 50 samples, as otherwise compute time is infeasible within the scope of this study). The average of the infidelities for each noise realisation is computed and returned as $\langle 1 - \mathcal{F}\rangle$ – this is the cost function we minimise to obtain control parameters for robust SDKs.

2.3. Numerical methods

Here, we describe the optimisation framework and search methods implemented in our code base for robust SDK design. A summary of the optimisation parameters is provided in Table 1.

2.3.1. Basic simulation code

Given our search space is $2M + 2$ (M pulse areas, M phases, dt and ω_A) dimensional, it is very unlikely that it contains just one minimum. Indeed, we see that the pulse unitary, Eq. (5), depends on a Bessel function and complex exponential function which introduce sinusoidal interference-like terms in the fidelity; therefore, as a result of a sinusoidal cost function, we expect to find many minima. To explore the search space more thoroughly we initialise the optimiser at forty-eight independent random seeds (reflecting the available CPUs). For each seed we perform a standard optimisation (without noise averaging) and record the best resulting infidelity. Only the best seed is then promoted to the noise-averaged optimisation stage, since the noise sampling is computationally intensive. This enables a quasi-global search while leveraging parallel execution across multiple CPUs.

For each initial guess, we also implement a convergence condition on the minimisation. In particular, for each output of the minimising operation, we check if the result is within 5% of the previous minimisation. In the case of the first minimisation, we compare to the initial guess. If the result is not within 5% of the previous minimisation, we take the control parameters and use them as our new initial guess and then check again if the result is within 5%. This process continues until the infidelity converges within 5%. The lowest converged infidelity across all initial seeds is subsequently chosen to be subjected to the robust search.

In terms of modelling the SDKs, we implement Eq. (5) and Eq. (6) and combine them as described by Eq. (7) to create an SDK. We then act this on the initial state, $|0, \alpha\rangle$ and compute the infidelity according to Eq. (8). The control parameters are then minimised whilst being subjected to noise to return a robust SDK sequence. For comparison, the code also allows for optimal SDKs, which are the case of no noise.

2.3.2. Simulation parameters and initial conditions

When designing our robust SDKs we are limited primarily by the following experimental constraints. As discussed in Section 2.1, we assume that our total SDK duration is orders of magnitude faster than the motional timescale of the ion requiring that our pulse timings are on the order of hundreds of picoseconds. Therefore, the minimum pulse separation needs to be resolvable by a high-repetition laser – our laser has a $\sim$ 3 GHz repetition rate. Additionally, the laser power is finite; however, the constraints imposed by this are minimal as the pulse sequence should at most do a π rotation so we do not require pulses with an area greater than π. Although we might be able to construct arbitrarily optimal robust SDKs, we need to impose bounds on the variables in our search space that also coincide with the achievable experimental constraints mentioned. These are summarised below in Table 1.

Table 1. Control parameters that are optimised to obtain low infidelity SDKs. The bounds require SDKs to be realised with physically achievable parameters.

Control Parameter	Pulse Area θ [rad]	Laser Phase ϕ [rad]	Beatnote ω_A [GHz]	Pulse separation dt [ns]
Bounds	$[0, \pi]$	$[-\pi, \pi]$	$[-2\pi, 2\pi]$	$[30 \times 10^{-3}, 10]$

We also note that the condition on the phase essentially allows for any laser phase to be chosen due to the periodicity of phase. The SDK beatnote, ω_A, bounds are such that they reflect previous experiments and are determined according to the hyperfine qubit transition frequency and the laser repetition frequency (Mizrahi et al. 2013a). This ensures that our solutions are physically meaningful and experimentally realisable SDKs may occur. It should be noted that pulse separation bounds reflect the laser that our collaborators use to implement our models.

3. RESULTS

3.1. State population

We begin by assessing the SDK's performance by confirming that the action of a numerically optimised SDK, with an infidelity of $1.51 \times 10^{-8} \pm 9.43 \times 10^{-11}$, acting on the initial state $|0, \alpha\rangle$ has the desired final state of $|1, \alpha + i\eta\rangle$.

This is shown in Figure 1 via a time evolution of the ion states using QuTip (Johansson et al. 2013) implemented in Python. Additionally, a bar chart showing the initial and final state populations for an unoptimised and optimised SDK are shown as a comparison.

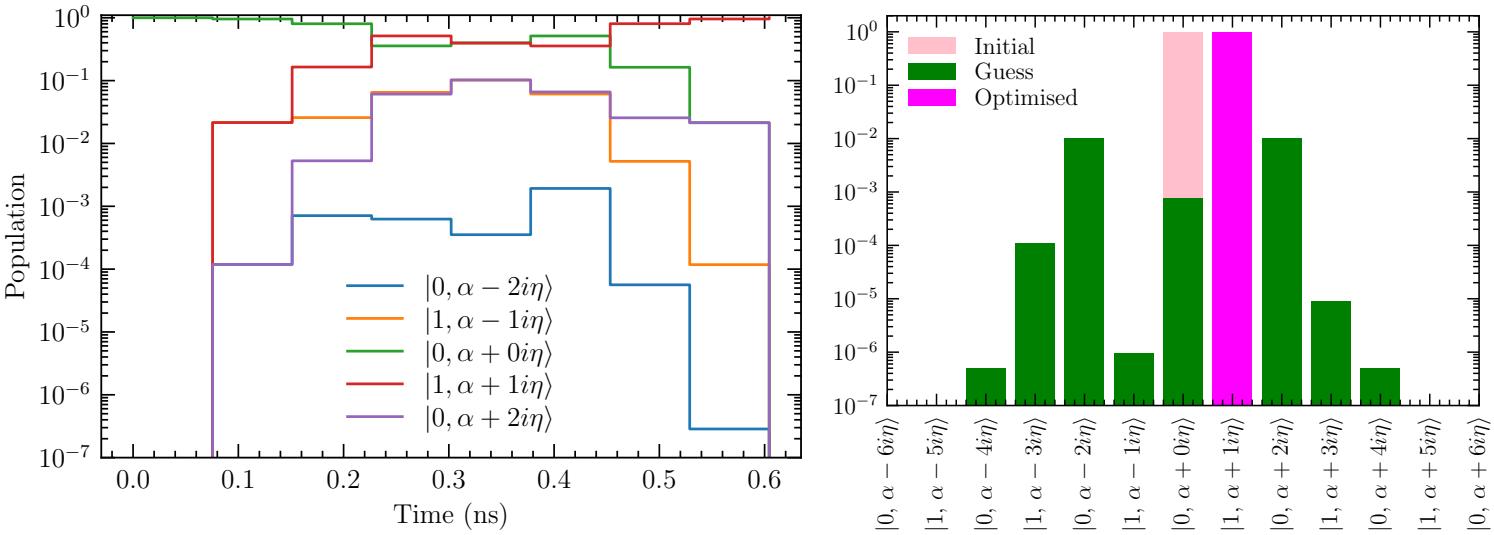


Figure 1. Time evolution and population distribution of a robust state-dependent kick (SDK) sequence. Time evolution of the population (left-hand panel) of each state during an 8-pulse robust SDK sequence. At $t = 0$, the ion is prepared in $|0, \alpha\rangle$. Each vertical step corresponds to the action of a pulse unitary, while horizontal segments represent free precession between pulses. The populations evolve such that only the target state $|1, \alpha + i\eta\rangle$ remains populated at the end of the sequence. Population distribution (left-hand panel) over all states after the SDK, comparing an unoptimised (random) initial guess and the optimised sequence. The unoptimised SDK shows population leakage into undesired states, whereas the optimised sequence suppresses unwanted transitions, yielding near-perfect fidelity in the target state. The optimisation is performed under an intensity noise threshold of 10^{-4}% RIN.

The left-hand panel of Figure 1 shows that at $t = 0$ the ion is completely in the state $|0, \alpha + 0i\eta\rangle$. We can identify vertical lines in the plot with a laser pulse, as $\hat{U}_\mathrm{p}$ acts to spread out the ion's motional states, and horizontal lines with the action of $\hat{U}_\mathrm{f}$ as there is no population change. Upon acting $\hat{U}_\mathrm{p}$ in the $\hat{U}_\mathrm{SDK}$ sequence, we observe a change in the population of this state, and also the appearance of other motional states of the ion. This produces a spread of motional coherences in phase space, and we require our SDK sequence to cancel out dispersion of the motional population by the end of the sequence. In the middle of the left-hand panel, we see that the population of these unwanted states is indeed considerable, on the order of 10^{-1}. However, the desired behaviour of our SDK is observed beyond this middle point as the populations of unwanted states quickly approach zero and the final state, $|1, \alpha + i\eta\rangle$, is maximally populated. This behaviour exemplifies the destructive interference mechanism described in Section 2.1. Importantly, this is only the case for an SDK sequence that uses optimised control parameters; for unoptimised parameters within the bounds of Table 1, multiple motional modes remain populated.

A similar visualisation is shown in the right-hand panel of Figure 1, where we show the result of an optimisation of control parameters on the final state. The ion starts in the fully populated state $|0, \alpha + 0i\eta\rangle$ and when acted upon an SDK with a randomised initial guess of control parameters, we clearly observe many unwanted populated states. This is significantly improved when the control parameters are optimised using our code infrastructure outlined in Section 2, with unwanted states cancelling out, leaving the desired final state populated maximally. Indeed, the final-state population increases from $\sim 10^{-2}$ to ~ 1 after optimisation. Moreover, we see that our truncation does not alter the physics as motional modes of $n > 5$ are not occupied in the final population. This highlights it is a suitable assumption for our model to include.

These results demonstrate that our optimisation routine is essential in designing robust SDKs with phase controls.

It should also be noted that the SDK in Figure 1 is optimised in the robust scheme with a noise threshold of 10^{-4} fractional RIN.

3.2. Control parameter optimisation

A key addition to our model is the consideration of the pulse phase. This allows an additional degree of freedom in the optimisation routine and should enable geometric cancellation of unwanted motional states. The inclusion of phase parameters does not degrade fidelity, as the optimiser can revert to the zero-phase case ($\phi_i = 0$) if beneficial. The phase-controlled scheme can therefore perform at least as well as, and generally better than, the phase-free case. A visualisation of the pulse areas and phases is shown in Figure 2.

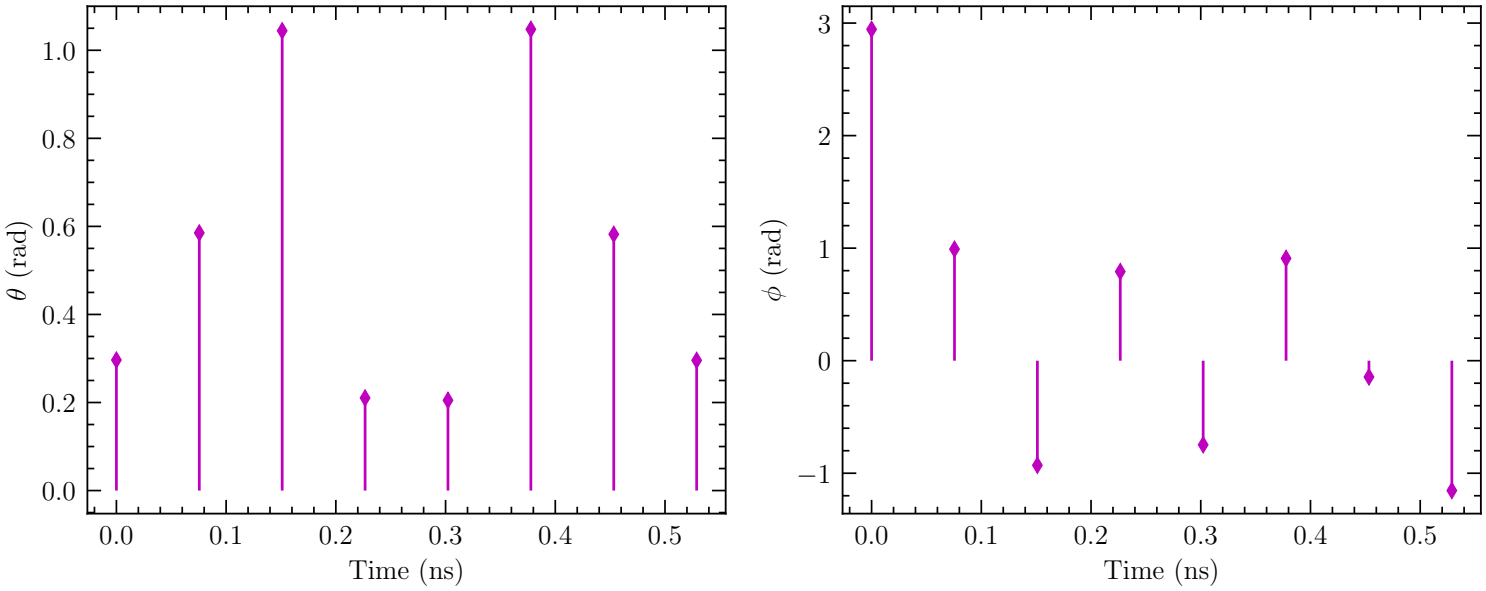


Figure 2. Optimised pulse areas (left-hand panel) in an 8-pulse SDK. The pulse areas show a symmetry about the middle of the pulse sequence. Optimised laser phases (right-hand panel) in an 8-pulse SDK. The laser phase is shown to contribute to the pulse sequence with contributions that are both positive and negative. Here, we use the same control parameters that were used to create Figure 1.

The left-hand panel of Figure 2 shows the pulse areas of an optimised phased controlled robust SDK. These describe the strength of each pulse in the SDK. While the magnitude of each pulse area is not of primary importance, it is noteworthy that the inclusion of the phase degree of freedom results in a noticeably different profile compared to the unphased case shown in Figure A1. In particular, the pulse area of the two middle pulses significantly reduced compared to the no-phase case. This is a consequence of the phase degrees of freedom increasing the dimension of the search space, and as such the optimiser identifies less intuitive yet higher-fidelity solutions.

The right-hand panel of Figure 2 highlights that the phases do indeed introduce complexity to the solutions. The non-trivial phases demonstrate that the optimiser exploits interference effects to geometrically cancel errors across the pulse train. This provides an additional, effective degree of freedom for constructing SDKs that are intrinsically robust against noise.

The timing between the pulses decreases by an order of magnitude ($\sim$ 1 ns to $\sim$ 0.1 ns) from the no-phase case (see Figure A1). Such a timing is experimentally realisable for our laser, meaning that this difference shows that gate durations can be overall reduced. This implies that the ratio of decoherence time to the duration of the gates can be further improved to three (from two) orders of magnitude compared to other leading platforms (Chen et al. 2014; Somoroff et al. 2023) (superconducting qubits) (Levine et al. 2019; Graham et al. 2022) (neutral atoms) whilst maintaining low infidelity SDKs.

3.3. Infidelity improvements

In Section 3.1 we visually saw the time evolution of the ion states populating the target final state, demonstrating that an optimised SDK presents the desired outcome. The phase degrees of freedom that we introduce in this paper were shown to have a non-zero, non-trivial contribution to the SDK in Section 3.2. Now, we turn our attention to measurable outcomes of our robust optimisation routine: its performance in the presence of noise compared to an optimal (noise free) SDK. In the following, we consider SDKs composed of 8 pulses, as these configurations balance gate duration with the number of pulses required to achieve a maximally populated final state.

3.3.1. 8−pulse sequence

Here, we consider an 8−pulse robust SDK sequence, which we compare to the optimal, noise-free solution. This is shown for both (fractional and absolute) RIN and (fractional) pulse timing noise, as these pose the greatest barriers in achieving low infidelities in our experiments.

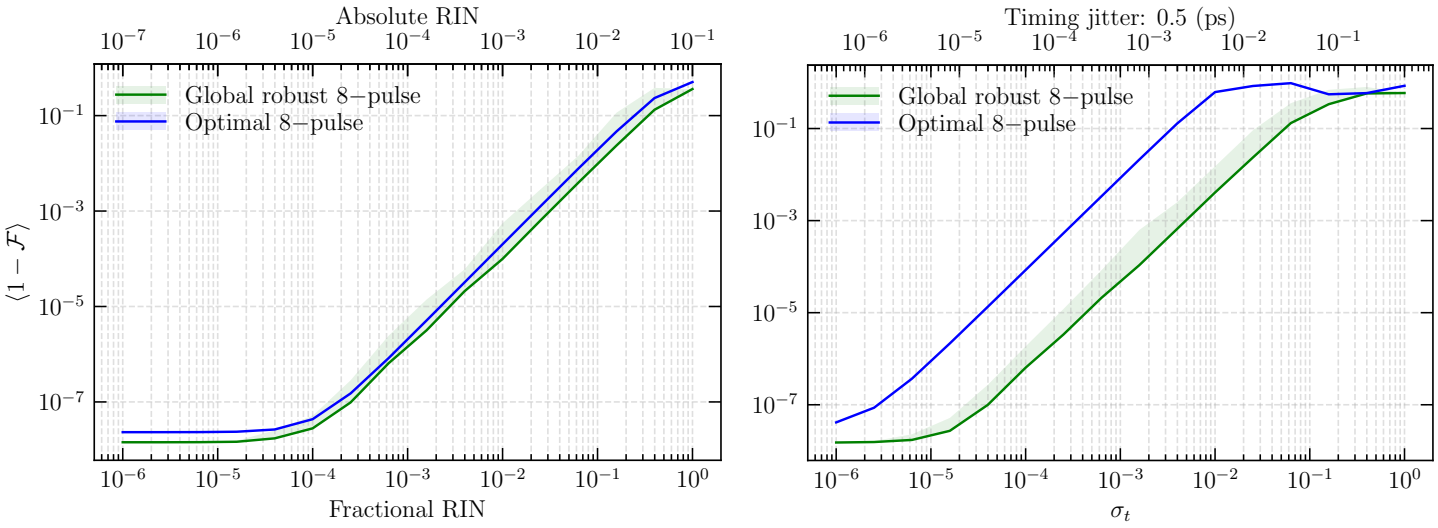


Figure 3. A 8−pulse robust and optimal SDK (left-hand panel) subject to random intensity noise (RIN) ranging from 10^{-4}% to 100% of the pulse area (bottom x-axis). The infidelity remains below 10^{-3} up to 0.01% RIN. The 'Optimal 8−pulse' label indicates an 8-pulse SDK with parameters that were optimised with no noise, to act as a baseline comparison. The label of 'Global robust 8−pulse' corresponds to a robust 8−pulse SDK. This was averaged over 50 noise realisations under a global search, as outlined in Section 2.3. The right-hand panel is identical to the left-hand with only the noise source now being the fractional timing jitter, σ_t, (bottom x-axis) between the pulses. The top x-axis shows the fractional timing jitter multiplied by an expected noise of 0.5 ps. The shaded region shows the uncertainty in the infidelity (due to the random Gaussian noise sampling) which is taken as the standard deviation over another 500 noise realisations.

We see that the mean fidelity for the robust solution consistently outperforms the optimal solution. This is seemingly counter-intuitive given that in the low noise regime ($< 10^{-4}$), the optimal solution should produce the lowest infidelity SDK. This reflects that the search space is highly complex and global searches are able to find marginally better solutions depending on its random seeds. As we study solutions that are designed to be robust (fractional RIN $> 10^4$), these variations of low noise where the infidelities are essentially invariant are not considered.

Instead, we focus on the regime in which the fractional RIN in Figure 3 is $> 10^{-4}$ where the robust solution leads to lower infidelities as compared to the optimal SDK sequence. Indeed, this is expected given that the optimal solution is found assuming no noise and so the quality of solution should degrade considerably when noise is introduced. However, the improvement reaches roughly a factor of three to five at 10^{-2} fractional RIN which, when considered in terms of the associated error bars, underperforms compared to the optimal solution. This demonstrates that finding robust solutions that significantly improve the infidelity compared to the optimal solution is challenging and suggests that alternative optimisation strategies may be required. The main difficulty when optimising under RIN is that the noise impacts $M = 8$ pulses, compounding this error such that the optimisation routine fails to find solutions with

a substantial gain. Future work may consider introducing more pulses and decreasing the bound on the pulse area bounds in Table 1 to $[0,\pi/2]$ so that each pulse suffers less RIN and the target state is still maintained.

In contrast, the robust optimisation scheme performs well compared to the optimal SDK with an infidelity improvement of 10^{-2} for all fractional timing noise in the range $10^{-4} < \sigma_t < 10^{-2}$. We also note that the timing jitter begins to impact the optimal solution about an order of magnitude before the robust solution, indicating a strong resilience to timing noise. The improvement that the robust scheme offers quickly deteriorates for large fractional timings. This should be expected given that with greater noise, the quality of timing in the free precession unitary (Eq. (6)) is reduced making it challenging for the SDK sequence to cancel out unwanted motional modes. Indeed, if we were to design an SDK sequence with more than eight pulses (as suggested in discussion of the left-hand panel) then this timing jitter would magnify, further worsening the infidelity of the sequence. This, therefore, imposes a restriction on the number of pulses which is reflected in our experiments.

Notably, the optimal solution lies beyond the one-sigma uncertainty band of the robust scheme. This highlights the strength of this approach for improving SDKs that can be operated under substantial timing jitter $\sigma_t = 10^{-3}$ with an infidelity two orders of magnitude better than the optimal case.

4. CONCLUSIONS

In this work, we used numerical simulations to study the implementation of phase as a parameter in designing robust SDKs for trapped-ion quantum computing. We modelled the effect of an ideal SDK (see Eq. (1)) using a combination of free precession and pulse unitaries (see Eqs. (5) and (6)) and optimised control parameters to perform under the presence of experimental noise. We observed that our SDK scheme leads to the correct final state population (Section 3.1), and that phases led to unintuitive solutions that act to cancel errors geometrically (Section 3.2) and infidelity improvements under timing jitter and RIN (Section 3.3). We summarise our key results below:

(i) We assessed the time evolution of the state population and observed the spreading out of motional states due to the action of our pulse unitary, Eq. (5). By the end of the sequence, unwanted motional states are suppressed which is achievable only with optimised control parameters.

(ii) A visualisation of the pulse areas and phases indicated that unintuitive solutions are formed due to the inclusion of phases in our model (see Figure 2). The inclusion of phases leads to non-trivial, unintuitive solutions that exploit interference to cancel errors.

(iii) Finally, we produce plots of noise averaged infidelity, $\langle 1 - \mathcal{F} \rangle$, and demonstrate that there is a significant (two orders of magnitude) gain in using our robust SDK scheme in the presence of timing jitter.

This work presents foundational results for phase-controlled SDKs in robust schemes that can be eventually implemented in full-scale trapped-ion systems. We also identify directions for future studies to investigate alternate methods for increasing the gap between the infidelity of robust schemes to the benchmarked optimal solutions. Additionally, this work sets the basis for investigating phase-controlled robust SDK schemes that operate under both RIN and timing jitter at the same time.

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APPENDIX A: PULSE AREAS OPTIMISED WITH NO PHASES

In this Appendix, we present unoptimised pulse area (left-hand panel) which is initially set to be a fixed value of $\pi/8$. The optimised pulses (right-hand panel) present a symmetric profile of pulse areas reflecting that when the phase is zero, the pulse areas are somewhat intuitive.

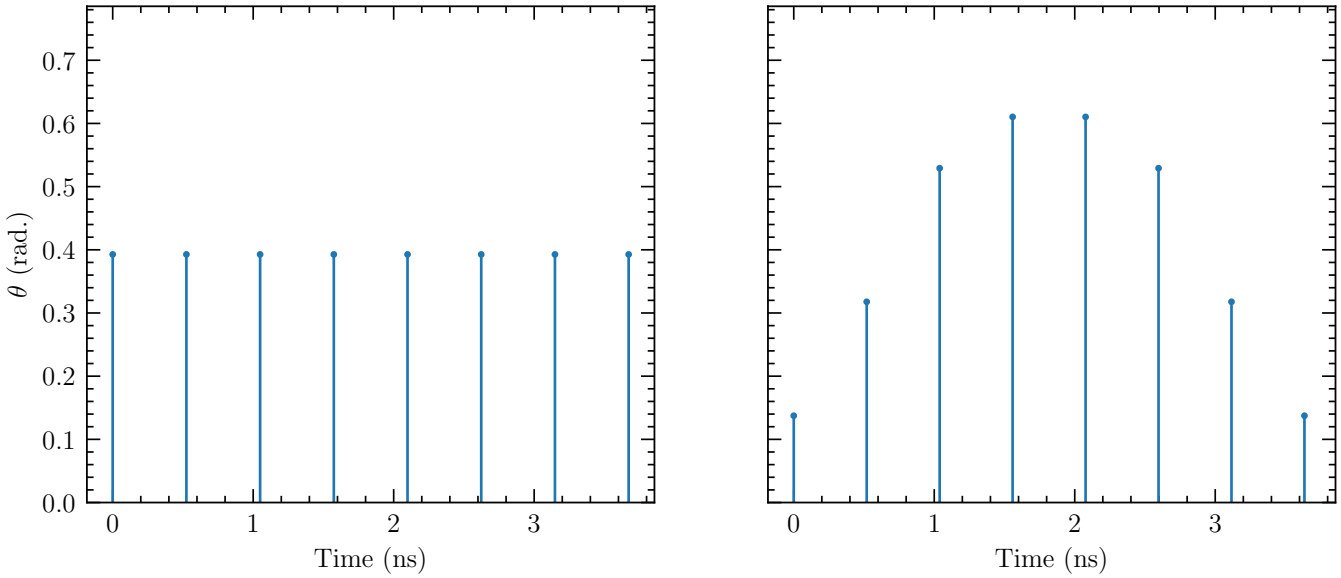


Figure A1. Initial uniform guess for pulse areas (left-hand panel). Optimised pulse areas (right-hand panel), with an SDK unitary that does not consider the phase of the laser.